HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and delivered **Power BI** and **Tableau** executive dashboards surfacing manufacturing KPIs and yield analytics for a global pharmaceutical manufacturer.
- Built and maintained **ETL/ELT pipelines** integrating data from **OSI PI**, **MES**, and **LIMS** systems into **SQL Server** and **Oracle data warehouses**.
- Optimised **SQL query structures** and **data models**, improving dashboard load times and ensuring reporting scalability for growing data volumes.
- Developed **Python automation** scripts for data reconciliation and exception reporting, reducing manual data handling by approximately **40%**.
- Implemented **GxP-compliant data integrity controls**, including automated validation rules and audit trails, adhering to FDA 21 CFR Part 11 guidelines.
- Served as the primary data liaison between manufacturing engineers, QA leads, and IT architects, translating operational requirements into scalable reporting solutions.
- Authored technical documentation for **ETL processes** and reporting assets in **Confluence**, supporting audit readiness and knowledge transfer for a regulated environment.

SKILLS

BI & Data Visualization – Power BI, Tableau, DAX, Power Query, Data Modelling

Databases & Data Warehousing – SQL Server, Oracle, PostgreSQL, MySQL, SQL

Data Engineering & ETL – ETL, ELT, Data Pipelines, Python, SQL

Manufacturing & Pharma Systems – OSI PI, PI Integrator, MES, LIMS, QMS, BIOVIA Discoverant

Cloud & Infrastructure – AWS, EC2, S3, Lambda, CloudWatch, Docker

Development & Automation – Python, Java, Spring Boot, REST APIs, Jenkins, Git, Linux

PROJECTS

GxP Compliant Manufacturing Analytics Platform

- Developed a suite of **Power BI** dashboards to visualise key manufacturing KPIs, including OEE and batch quality metrics, for operational teams and senior leadership.
- Engineered **ETL pipelines** using **Python** and **SQL** to ingest data from disparate manufacturing systems (MES, LIMS) into a central **SQL Server** data warehouse, ensuring data integrity.
- Implemented automated data validation checks and comprehensive audit trails within the data pipelines to meet **GxP** and 21 CFR Part 11 requirements.

Real-time Production Monitoring System

- Built **SQL** queries and stored procedures to extract and transform real-time data from an **OSI PI** historian for performance analysis.
- Created interactive **Tableau** visualisations to present equipment performance and yield trends, enabling faster identification of production bottlenecks.
- Automated data reconciliation processes using **Python scripting**, reducing manual effort by approximately **30%** and improving reporting accuracy.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)